

House Price Prediction — Executive Summary

Internship Project – Week 1

Dataset: Kaggle Housing Prices Dataset (545 Residential Properties)

Project Objective

The primary objective of this project was to develop a **machine learning model** capable of accurately estimating residential property prices based on structural characteristics and housing amenities while identifying the features that contribute most significantly to market value.

Beyond prediction, the project aimed to derive **data-driven business insights** that could support better pricing strategies and real estate decision-making.

Data Preparation & Preprocessing

The dataset consists of **545 residential properties** containing **12 predictive features** and **one target variable (price)**. An initial exploratory analysis confirmed that:

- No missing values were present.
- No duplicate records were found.
- The dataset was clean and ready for modelling.

To prepare the data:

- Binary categorical variables (yes/no) were converted into numerical values (1/0).
- The multi-category *furnishingstatus* attribute was transformed using **One-Hot Encoding**.
- The processed dataset was divided into **80% training set** and **20% testing set** for unbiased model evaluation.

Model Development & Evaluation

Two supervised learning algorithms were implemented and evaluated on the test dataset to compare their predictive performance:

Model	MAE (₹)	RMSE (₹)	R ² Score
Linear Regression	970,043	1,324,507	0.6529
Random Forest Regressor	1,022,545	1,403,779	0.6101

Observation

Interestingly, the **Linear Regression model outperformed the Random Forest Regressor across all evaluation metrics**.

Given the relatively small dataset size and predominantly linear relationships among variables, the simpler model generalized more effectively and produced lower prediction errors.

The Linear Regression model explains approximately **65% of the variation in housing prices**, demonstrating reasonable predictive capability for practical applications.

Key Findings

The analysis revealed several important insights:

- **Property area** is the strongest determinant of house price.
- **Number of bathrooms** has the second-highest influence on valuation.

- **Air conditioning** and **additional stories** positively impact property prices.
- Surprisingly, **bedroom count contributes less to price prediction than expected**, suggesting buyers value usable living space and amenities more than simply having additional rooms.

Business Recommendation

Based on the analysis, real estate organizations should prioritize:

- Accurate measurement and presentation of **property area**.
- Highlighting premium amenities such as **bathrooms and air conditioning**.
- Leveraging **interpretable linear pricing models** for automated valuation systems, particularly when working with structured datasets of similar size.

Such an approach provides both **high predictive performance** and **greater transparency for business stakeholders**.

Future Scope

Although the Linear Regression model performed well, future improvements may include:

- Hyperparameter optimization
- Advanced feature engineering
- Feature selection techniques
- Ensemble boosting algorithms such as **XGBoost, LightGBM, or CatBoost**
- Cross-validation for improved model robustness and generalization

These enhancements could potentially increase predictive accuracy and make the pricing system more scalable for larger real-world datasets.

Conclusion

This project demonstrates how **data preprocessing, exploratory analysis, feature engineering, and supervised machine learning** can be effectively combined to solve a practical real estate valuation problem.

An important takeaway is that **model complexity does not always guarantee better performance**. In this case, a simple and interpretable **Linear Regression model** proved more effective than a more sophisticated ensemble approach.

This project strengthened my understanding of the complete machine learning lifecycle—from data preprocessing and feature engineering to model development, evaluation, and interpretation—while reinforcing the importance of evidence-based decision-making in practical data science applications.